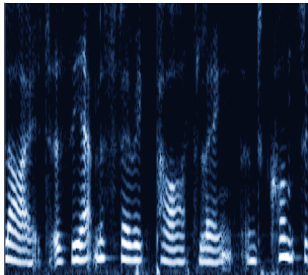
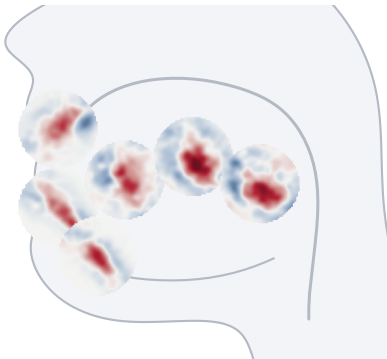


Where the mouth stops reaching

Ordinary voice recording



deep
learning



In Parkinson's disease

Tongue & jaw move less

up to 1.0 SD, 2 languages

Tracks speech severity

$\rho = -0.87$ (UPDRS II)

Jaw tremor at 3-7 Hz

separate from voice tremor

Validated on real sensors

adds to classic acoustics